

Listing 9.9. Mean field Q-learning: The policy function

```
from collections import deque
1
from random import shuffle
2

def softmax_policy(qvals,temp=0.9):
3
    soft = torch.exp(qvals/temp) / torch.sum(torch.exp(qvals/temp))
4
    action = torch.multinomial(soft,1)
5
    return action
```

- ***1*** We will use the deque data structure as an experience replay storage list, since it can be set to have a maximum size.
- ***2*** We will use the shuffle function to shuffle the experience replay buffer.
- ***3*** This policy function takes in a Q value vector and returns an action, either 0 (down) or 1 (up).
- ***4*** This is the softmax function definition.
- ***5*** The softmax function converts the Q values into a probability distribution over the actions. We use the multinomial function to randomly select an action weighted by the probabilities.

The first new function we're going to use for the 2D Ising model is the softmax function. You saw this before in chapter 2 when we introduced the idea of a policy function. A policy function is a function, π:$S \rightarrow A$, from the